Thanh Van Trong Tran

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Education

Nanyang Technological University (NTU) — Singapore
Ph.D. in Electrical and Electronic Engineering — *Aug 2026 – Present*
- Advisor: Prof. Woon-Seng Gan

VNU University of Engineering and Technology (VNU-UET) — Hanoi, Vietnam
Bachelor in Computer Science (Honors) — *Aug 2019 – Jun 2023*
- Thesis: Automatic Speech Recognition in Low-Resource Condition
- GPA: 3.71/4.00 (top 5%)

Experience

AI Research Resident — Aug 2023 – Jul 2026
FPT Software – AI Center — *Hanoi, Vietnam*
- Researched speech AI and multimodal learning for emotion recognition, vision-guided audio and speech synthesis.
- Developed black-box optimization methods in discrete and latent spaces to improve protein sequences.
- Supervisor: Prof. Truong-Son Hy and Dr. Van Nguyen.

Student Researcher — Sep 2022 – May 2023
Institute for Artificial Intelligence (IAI), VNU-UET — *Hanoi, Vietnam*
- Researched Person/Vehicle Re-identification (ReID) and Instance Retrieval problems.
- Developed an ESPNet-like speech processing toolkit in TensorFlow 2, streamlining development and deployment.
- Supervisor: Dr. Long Quoc Tran

Speech Processing Intern — Oct 2021 – Nov 2022
VinBigData – Spoken Language Processing Department — *Hanoi, Vietnam*
- Designed a C-based telephony speech augmentation pipeline, reducing WER by 25% on production benchmarks.
- Developed a neural decoder that improved decoding accuracy by 10% over vanilla beam search.
- Delivered a low-latency streaming ASR model ($\sim$ 70ms) with approx. 10% WER for in-car assistant.
- Optimized the in-house ASR framework, improving scalability and processing performance.
- Developed a specialized QA app for the VinFast's Virtual Assistant, accelerating the EV deployment pipeline.

Selected Publications (Google Scholar)

Precise Video-to-Audio Generation with Cross-Modal Alignment in Latent Space
Thanh V. T. Tran, Ngoc-Son Nguyen, Luong Tran, Long-Khanh Pham, Paarth Neekhara, Shehzeen Hussain, Van Nguyen
European Conference on Computer Vision (ECCV), 2026. [paper] [code] [website]

DiFlow-TTS: Compact and Low-Latency Zero-Shot Text-to-Speech with Factorized Discrete Flow Matching
Ngoc-Son Nguyen, Thanh V. T. Tran, Hieu-Nghia Huynh-Nguyen, Truong-Son Hy, Van Nguyen
Interspeech (Long Paper track), 2026. [paper] [code] [website]

DiFlowDubber: Discrete Flow Matching for Automated Video Dubbing via Cross-Modal Alignment and Synchronization
Ngoc-Son Nguyen, Thanh V. T. Tran, Jeongsoo Choi, Hieu-Nghia Huynh-Nguyen, Truong-Son Hy, Van Nguyen
Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026. [paper] [code] [website]

Speech Reconstruction from Silent Videos via Acoustic-Semantic Decomposed Modeling
Long-Khanh Pham, Thanh V. T. Tran, Minh-Tan Pham, Van Nguyen
Interspeech, 2025. [paper] [code] [website]

Effective Context Modeling Framework for Emotion Recognition in Conversations
Cuong Tran Van, Thanh V. T. Tran*, Van Nguyen, Truong-Son Hy*
International Conference on Acoustics, Speech, and Signal Processing (ICASSP), 2025. [paper] [code] [poster]

GROOT: Effective Design of Biological Sequences with Limited Experimental Data
Thanh V. T. Tran, Nhat Khang Ngo*, Viet Anh Nguyen, Truong-Son Hy*
Conference on Knowledge Discovery and Data Mining (KDD), 2025. [paper] [code] [poster] [slides]

Honors and Awards

The Rising Innovator Apr 2026
FPT Software – AI Center
- Achieved *Third Place* among 20 finalists in the internal AI Innovation Hackathon Q1 2026.

Best Performance Award 2025 Jan 2026
FPT Software – AI Center
- Recognized as one of the top 3 performers among 100+ AI engineers and researchers in 2025.

Second Prize 2022
Association for Vietnamese Language and Speech Processing
- Achieved *Second Place* in Task 1: Automatic Speech Recognition at VLSP 2022, surpassing 30+ teams nationwide

Third Prize 2022
Association for Vietnamese Language and Speech Processing
- Achieved *Third Place* in Task 2: Automatic Speech Recognition at VLSP 2022, surpassing 30+ teams nationwide.

Skills

Languages: Vietnamese (Native), English (IELTS 7.5)
Programming and Software: Python, C/C++, Bash, Git, LaTeX, Docker, Linux
Machine Learning Package: PyTorch, TensorFlow, Transformers, Scikit-Learn, NVIDIA-NeMo

Academic Services

Reviewer
ACMMM 2025, ICASSP 2027

Activities

Open Source Contributor

- **NVIDIA-NeMo/Speech:** Fixed timestamp computation logic in streaming multi-talker ASR for the Parakeet model (#15701).

Data Science Mentor 2023
Math and Summer Science Program (MaSSP)
- Lecturer at MaSSP (a non-profit summer STEM program for Vietnamese high school students), teaching Calculus and Foundations of Computer Vision.